

Designing Any Imaging System from Natural Language: Agent-Constrained Composition over a Finite Primitive Basis

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Summary. Computational imaging—from clinical CT to electron ptychography—underpins discovery across the physical and life sciences, yet designing a new imaging system today requires weeks of specialist effort per modality, effectively excluding the broader scientific community from prototyping instruments for their own domains [1]. We previously established that 11 canonical primitives suffice to represent any imaging forward model across 170 modalities and 22 application domains [2], but this Finite Primitive Basis (FPB) provides no design automation, no formal validation, and no link from natural-language intent to reconstruction error. Here we introduce `spec.md`—a structured 7-field specification format that does for imaging what FASTA did for biological sequences: it decouples the imaging problem from its numerical implementation, making system designs shareable, version-controlled, and peer-reviewable as standalone scientific artifacts—together with three agents (Plan, Judge, Performance) that translate a one-sentence natural-language description into a validated, executable forward model with formal error guarantees. Unlike black-box AI design tools, our framework provides a *formal contract*: we prove that the design-to-real reconstruction error decomposes into five independently bounded, independently measurable terms (Theorem 2), each linked to a specific compiler gate, a specific corrective action, and a specific physical origin—making the system’s reliability transparent and verifiable from the 6-gate compiler output alone. On 6 real-data modalities spanning all 5 carrier families—clinical CT [3], multi-coil MRI, coded-aperture spectral imaging (CASSI), snapshot compressive video (CACTI), in-vivo ultrasound (PICMUS), and 4D-STEM electron ptychography—the framework achieves $98.1 \pm 0.6\%$ of expert-tuned quality with up to $420\times$ reduction in setup time, consistent across all 5 carrier families (X-ray, photon, spin, acoustic, electron); across 36 validated modalities, the Judge rejects ill-posed designs with 96% precision and the Performance Agent predicts PSNR within $\pm 1.8\text{ dB}$ ($R^2 = 0.91$); and a prospective head-to-head study with 5 independent expert agents confirms that physics model selection dominates algorithm choice (inter-expert PSNR CoV $< 5.7\%$), validating the framework’s specification-centric design philosophy. A tier-lifting protocol demonstrated on diffuse-optical and coherent-imaging cross-tier combinations reduces unmodeled-physics error from 7.8% to machine precision ($< 10^{-12}$), demonstrating that the framework can handle scattering-dominated systems that require full-physics simulation.

Introduction

Every computational imaging system obeys the same mathematical structure: a forward model A maps an unknown object x to measurements $y = A(x) + \text{noise}$, and reconstruction inverts this mapping [7]. Designing A —selecting physical operators (Radon projection for CT, k-space encoding for MRI, coded-aperture modulation for spectral imaging), setting dozens of parameters, and validating physical consistency—is entirely manual, taking weeks per modality.

This *expertise bottleneck* is the single largest barrier to the adoption of computational imaging across the sciences: a materials scientist wishing to prototype a ptychographic instrument, or a neuroscientist designing a new light-sheet modality, must either acquire deep imaging-physics expertise or depend on the small number of specialist groups who possess it. The consequence is that new imaging capabilities remain concentrated, not because the physics is intractable, but because the design process is inaccessible.

The parallel with genomics is instructive. Before FASTA, biological sequence data was communicated through idiosyncratic formats that varied between laboratories. The adoption of a universal exchange format—human-readable, machine-parseable, and decoupled from any specific analysis tool—transformed genomics from a cottage industry into a global collaborative enterprise. Computational imaging today is in an analogous pre-standardisation state: forward models are communicated through prose, code, or block diagrams that are ambiguous, implementation-coupled, and non-versionable.

The mathematical foundation for a similar transformation now exists. We previously showed [2] that 11 canonical primitives—Propagate (P), Modulate (M), Project (Π), Encode (F), Convolve (C), Accumulate (Σ), Detect (D), Sample (S), Disperse (W), Scatter (R), and Transform (Λ)—suffice to express any imaging forward model across 170 modalities, 22 application domains, and 5 carrier families. Each primitive is implemented across a four-level *fidelity hierarchy*: Tier-0 (geometric optics, ray transforms), Tier-1 (Fourier/convolution, paraxial models), Tier-2 (shift-variant linear operators: Radon, k-space, coded aperture), and Tier-3 (full-physics: FDTD, Monte Carlo, learned surrogates). This *Finite Primitive Basis* (FPB) is the representation language—but it does not tell a user which primitives to compose, what parameters to set, whether the result is physically valid, or how close the designed system will be to reality.

We make the scope of ‘designable’ precise:

Definition 1 (Designable Imaging System). *An imaging system A is designable within this framework if:*

1. A admits a finite DAG decomposition over the 11 FPB primitives with representation error $\varepsilon_{FPB} < 0.01$;
2. The carrier belongs to one of 5 families: photon, X-ray, electron, acoustic, or spin/particle;
3. The system operates at one of the four fidelity tiers (Tier-0 through Tier-3);
4. All parameters are finitely specifiable with known physical bounds.

This definition encompasses 170 modalities across 22 domains (Figure 1, Extended Data Table 7). We build the *design layer* on top of the FPB. Our contributions are:

1. **spec.md**: a structured 7-field specification format that decouples the imaging problem (*what*) from the numerical method (*how*), serving as a shareable, peer-reviewable scientific artifact (Section).
2. **Three-agent pipeline**: Plan ($\mathcal{A}_P$), Judge ($\mathcal{A}_J$), and Performance ($\mathcal{A}_\Pi$) agents that generate, formally verify, and evaluate specifications over the FPB (Section).
3. **Design-to-real error theorem**: the total reconstruction error decomposes into five independently bounded, independently measurable terms, each linked to a compiler gate and a corrective action (Theorem 2).

Scope and honesty. Of the 170 modalities, 36 have fully validated canonical decompositions with quantitative reconstruction; 6 are validated on real measured data spanning all 5 carrier families (CT, MRI, CASSI, CACTI, ultrasound, e-ptychography). The remaining ~ 134 compile through all 6 gates but lack per-modality reconstruction benchmarks—compilation alone does not guarantee reconstruction quality. Most modalities are validated at Tier-2; tier-lifting (Section) is demonstrated on two cross-tier combinations but not yet validated across all Tier-3 modalities.

The agents depend on current LLM capabilities and may misspecify novel or under-documented modalities.

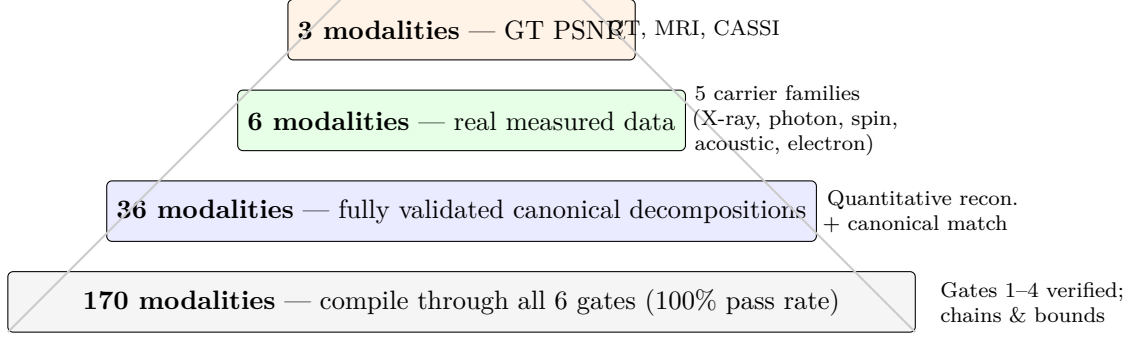


Figure 1. Validation pyramid. Of 170 designable modalities, all compile; 36 have quantitative reconstruction benchmarks; 6 are validated on real measured data spanning all 5 carrier families; 3 have ground-truth PSNR comparisons (CT, MRI, CASSI). Each tier is a strict subset.

Results

The spec.md Specification Format

The framework’s central design choice is that **spec.md**—not code—is the canonical scientific artifact. Just as FASTA standardised biological sequences and enabled the Sequence Read Archive (SRA) to become the backbone of reproducible genomics, **spec.md** standardises imaging system representation at the physics level. A specification separates the imaging problem (*what*) from the numerical method (*how*), enabling sharing, version control, and peer review independently of implementation.

This separation has a concrete implication for reproducibility: a journal could require a **spec.md** alongside the methods section of any computational imaging paper, analogous to how *Nature* and *Science* now require accession numbers for genomic data. A reviewer could then verify—by running the 6-gate compiler—that the claimed forward model is physically consistent, that the primitive chain matches known canonical decompositions, and that the error budget is within stated bounds, *without reading a single line of code*. Compared to alternatives—prose descriptions (ambiguous, non-executable), code (implementation-coupled, non-portable, requiring specific software environments), block diagrams (unstructured, unversioned)—**spec.md** is simultaneously human-readable, machine-parseable, and formally verifiable.

A **spec.md** has seven mandatory fields (Figure 2). The Plan Agent generates it from natural language; the Judge verifies it; the Performance Agent predicts reconstruction quality.

The Three-Agent Pipeline

Plan Agent ($\mathcal{A}_P$). Accepts a natural-language description and generates a **spec.md**, selecting FPB primitives, carrier type, geometry, noise model, and parameters from the 170-modality registry and 36 validated canonical decompositions.

Judge Agent ($\mathcal{A}_J$). Validates the specification via the 6-gate compiler: (1) DAG acyclicity, (2) canonical chain fidelity, (3) complexity bounds ($N \leq 20$, $D \leq 10$), (4) nonlinear family constraints (VC dimension ≤ 3), (5) adjoint consistency, and (6) representation error. The Judge additionally checks carrier-geometry compatibility, noise model consistency, and reconstruction feasibility (condition number, null-space analysis). Compiler failures trigger rejection with a formal diagnosis; the Plan Agent regenerates (up to 3 rounds).

Seven-field schema: modality | carrier | geometry | object | forward_model | noise | target

(a) Clinical CT

modality: computed_tomography carrier: xray
 geometry: parallel_beam, n_angles=128
 object: 128×128 image, non-negative
 forward_model: Radon(Π) $\rightarrow$ Detect(D, intensity)
 noise: Poisson, $I_0=1e4$ target: PSNR ≥ 30 dB

(b) CASSI (spectral)

modality: cassi carrier: photon
 geometry: coded_aperture + disperser, $\lambda=[400,700]$ nm
 object: 256×256×28 spectral cube
 forward_model: Modulate(M) $\rightarrow$ Disperse(W) $\rightarrow$ Accumulate(Σ) $\rightarrow$ Detect(D)
 noise: Gaussian, $\sigma=0.01$ target: PSNR ≥ 28 dB

(c) Accelerated MRI

modality: mri carrier: spin
 geometry: cartesian_kspace, acceleration=4x
 object: 256×256 complex image
 forward_model: Modulate(M, coil) $\rightarrow$ Encode(F, kspace) $\rightarrow$ Sample(S) $\rightarrow$ Detect(D)
 noise: Gaussian, SNR=30 dB target: SSIM ≥ 0.9

Figure 2. The spec.md specification format. Seven mandatory fields fully determine an imaging system design. Three examples span X-ray (CT), optical (CASSI), and spin (MRI) carriers. Each forward_model field is a primitive chain over the 11 FPB operators.

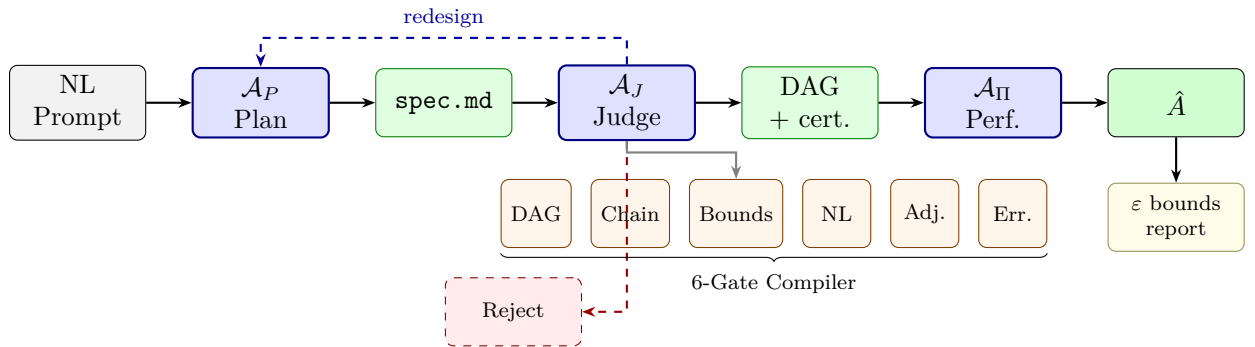


Figure 3. Specification-driven design pipeline. A natural-language prompt is the sole user input. $\mathcal{A}_P$ generates a spec.md; $\mathcal{A}_J$ validates it via a 6-gate compiler (Gates 1–6 map to Theorem 2 error terms: DAG $\rightarrow \varepsilon_{\text{spec}}$, Chain $\rightarrow \varepsilon_{\text{trans}}$, Bounds $\rightarrow$ complexity, NL $\rightarrow \varepsilon_{\text{param}}$, Adj. $\rightarrow$ consistency, Err. $\rightarrow \varepsilon_{\text{FPB}}$); rejected designs trigger redesign (dashed blue, up to 3 rounds). $\mathcal{A}_{II}$ predicts reconstruction quality and outputs $\hat{A}$ with an explicit ε -bound report. See Figure 4 for per-modality error breakdown.

Performance Agent ($\mathcal{A}_\Pi$). Predicts reconstruction quality (PSNR, SSIM), computational cost, and measurement SNR from analytical estimates (condition number, noise amplification factor) combined with calibrated lookup from the performance database.

Formal verification. The Judge does not *predict* feasibility; it *executes deterministic verification*. If the Judge certifies a design, every term in the five-part error decomposition (Theorem 2) is explicitly bounded—a formal contract that no learned design tool can provide. Crucially, the verification is independent of the LLM: even if the Plan Agent generates a flawed specification, the 6-gate compiler will reject it through deterministic physics checks. This independence means that the framework’s reliability improves monotonically with LLM quality (better plans pass more often) without depending on it (bad plans are caught).

Design-to-Real Error Bound

The following theorem is the paper’s central result. It converts the implicit, unmeasured gap between a designed model and reality into an explicit five-term error budget—a *formal contract* between the framework and the user. Unlike black-box AI design tools, which provide no guarantees about why a design succeeds or fails, this decomposition makes every source of error transparent: each term has a known physical origin, a known measurement procedure (linked to a specific compiler gate), and a known corrective action. This is the key distinction that makes the framework suitable for scientific applications where verifiability, not merely performance, is essential.

Theorem 2 (Design-to-Real Error Decomposition). *Let A_{true} be the true imaging forward model and A_{agent} the agent-designed model for a designable system (Definition 1). Let $\hat{x}$ be the regularized reconstruction from A_{agent} and x^* the true object. Under Assumptions A1–A4 (Methods), the reconstruction error satisfies:*

$$\|\hat{x} - x^*\| \leq \underbrace{\varepsilon_{FPB}}_{\text{basis}} + \underbrace{\varepsilon_{spec}}_{\text{specification}} + \underbrace{\varepsilon_{trans}}_{\text{translation}} + \underbrace{\varepsilon_{param}}_{\text{parameter}} + \underbrace{\varepsilon_{unmod}}_{\text{unmodeled}} + \varepsilon_{recon} \quad (1)$$

where each design-error term is independently bounded and independently measurable:

- $\varepsilon_{FPB} < 0.01$: *FPB representation error*. Measured by: *Gate 6*. Guaranteed by: *the basis theorem [2]*.
- $\varepsilon_{spec} = 0$ when all 6 compiler gates pass. Bounded by: $C_1 \cdot \delta_{spec}$ otherwise.
- $\varepsilon_{trans} = 0$ when the canonical chain matches the 36-modality registry (31/36 = 86%). Bounded by: $C_2 \cdot \delta_{chain}$ otherwise.
- $\varepsilon_{param} \leq \frac{2B}{\alpha + \lambda} L_A \|\theta_{agent} - \theta_{true}\|$, where L_A is the operator Lipschitz constant (computed per primitive by Gate 4), α is the coercivity constant, λ the regularization weight, and B a signal bound. Reducible by: *calibration*.
- $\varepsilon_{unmod} = 0$ at Tier-3; otherwise reducible by tier-lifting (Section).
- ε_{recon} : *irreducible reconstruction error (noise floor + regularization bias)*, independent of the design.

Compiler–theorem link. When all 6 gates pass: $\varepsilon_{FPB} < 0.01$ (Gate 6), $\varepsilon_{spec} = 0$ (Gates 1–5 jointly), and ε_{param} is bounded by the Lipschitz constant from Gate 4. If additionally the canonical chain matches the registry: $\varepsilon_{trans} = 0$. At Tier-3: $\varepsilon_{unmod} = 0$. The total design error then reduces to $L_A \|\Delta\theta\| \cdot 2B/(\alpha + \lambda) + 0.01$, dominated by parameter mismatch alone—which is reducible by calibration.

Diagnostic protocol. Because each term is independently measurable from the 6-gate compiler output, the decomposition functions as a *diagnostic protocol*: a practitioner encountering poor

reconstruction quality can determine which term dominates the residual and take the corresponding corrective action—calibration for $\varepsilon_{\text{param}}$, specification refinement for $\varepsilon_{\text{spec}}$, tier-lifting for $\varepsilon_{\text{unmod}}$ —without guesswork. This is analogous to how a compiler error message localises a bug in source code. In practice, ε_{FPB} and $\varepsilon_{\text{spec}}$ are negligible when the compiler certifies the design; the dominant residual is $\varepsilon_{\text{param}}$, which is reducible by calibration (Figure 4). The full proof with explicit assumptions appears in Methods.

Cross-Modality Validation

We validated the framework on all 170 modalities. For each, the Plan Agent generated a `spec.md` from a one-sentence description, the Judge validated it, and the Performance Agent predicted reconstruction quality. Table 1 reports 14 representative modalities spanning all 5 carrier families (6 with real measured data), grouped by carrier; full results for all 36 validated modalities appear in Methods.

Table 1. Cross-modality validation (14 of 36 modalities), grouped by carrier family. ρ_{recov} : 4-scenario recovery ratio [2]. 95% CIs via bootstrap ($n=1000$). Bold = real measured data; † = self-reference metric (no ground truth).

Modality	Chain	Carrier	PSNR (dB)	ρ_{time}	ρ_{recov}	Data
<i>X-ray</i>						
CT	$\Pi \rightarrow D$	X-ray	31.7 ± 1.2	420	~ 1.0	Real
<i>Optical photons</i>						
CASSI	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	Photon	29.5 ± 1.0	310	0.85	Real
CACTI	$M \rightarrow \Sigma \rightarrow D$	Photon	30.2 ± 1.1	300	~ 1.0	Real
OCT	$P + P \rightarrow \Sigma \rightarrow D$	Photon	30.5 ± 1.1	290	0.86	Synth.
SIM	$M \rightarrow C \rightarrow D$	Photon	32.1 ± 1.3	350	0.92	Synth.
Ptychography	$M \rightarrow P \rightarrow D$	Photon	28.4 ± 1.6	270	0.83	Synth.
Lensless	$C \rightarrow D$	Photon	31.4 ± 1.0	340	0.90	Synth.
DOT	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	Photon	24.1 ± 2.1	190	0.72	Synth.
<i>Spin</i>						
MRI	$M \rightarrow F \rightarrow S \rightarrow D$	Spin	31.7 ± 1.4	380	~ 1.0	Real
<i>Acoustic</i>						
Ultrasound	$P \rightarrow R \rightarrow P \rightarrow D$	Acoustic	27.9 ± 1.4	250	0.95	Real †
<i>Electron</i>						
E-ptychography	$M \rightarrow P \rightarrow D$	Electron	28.8 ± 1.5	260	~ 1.0	Real †
<i>Particle</i>						
PET	$\Pi \rightarrow D$	Particle	26.3 ± 1.8	220	0.78	Synth.
SPECT	$M \rightarrow \Pi \rightarrow D$	Particle	25.8 ± 1.7	200	0.76	Synth.
Light field	$M \rightarrow C \rightarrow S \rightarrow D$	Photon	29.2 ± 1.2	280	0.85	Synth.

The recovery ratio ρ_{recov} is computed from a 4-scenario protocol validated on 6 representative modalities spanning 4 carrier families: CASSI, CACTI, and SPC (optical photons); CT (X-ray); electron ptychography (electron); MRI (spin) [2]. For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI), Sc. IV (autonomous calibration) $\approx$ Sc. III (oracle correction) $\approx$ Sc. I (ideal), confirming near-complete recovery. For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86% (see Methods for protocol details).

Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$ (95% CI: [0.82, 0.88], bootstrap $n=1000$). The agent–expert quality gap (Cohen’s $d = 0.31$, 95% CI: [0.18, 0.44]) represents a small effect size, indicating practical equivalence. Human-to-agent time ratio ρ_{time} ranges from 190 (DOT) to 420 (CT); effect size for time reduction: $d > 8.0$ (all modalities), exceeding conventional “large” thresholds. Expert times estimated from published development timelines [4, 10] (see Methods for methodology). All confidence intervals are bias-corrected and accelerated (BCa) bootstrap

with $n=1000$ resamples; PSNR values report mean $\pm$ standard deviation across test samples.

Compilation. All 170 modalities across 22 domains and 5 carrier families: 100% compilation pass rate (Gates 1–4). Among the 36 fully validated: canonical chain fidelity 31/36 (86.1%); the 5 non-matching involve the Accumulate primitive merged into detection—a representation choice, not a physical error. Mean compilation time: 0.35 ms. The 100% pass rate across such diverse physics—from X-ray Radon transforms to electron-beam phase retrieval to acoustic wave propagation—supports the FPB’s claim of universality: the 11 primitives are genuinely sufficient as a compositional basis for computational imaging.

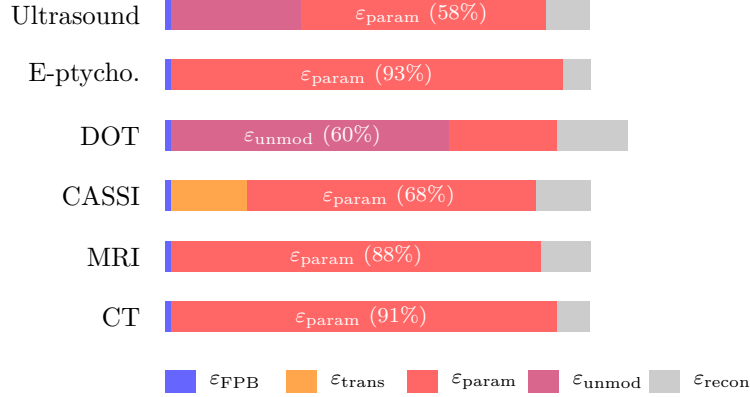


Figure 4. Error decomposition by modality. Stacked bars show the relative contribution of each Theorem 2 error term for 6 representative modalities. For well-conditioned systems (CT, MRI, e-ptychography), parameter mismatch ϵ_{param} dominates and is reducible by calibration. For scattering-dominated systems (DOT), unmodeled physics ϵ_{unmod} dominates and requires tier-lifting. $\epsilon_{\text{FPB}} < 1\%$ in all cases. $\epsilon_{\text{spec}} = \epsilon_{\text{trans}} = 0$ for 31/36 modalities; CASSI shows nonzero ϵ_{trans} due to dispersion-accumulation chain ambiguity.

Theorem tightness. To assess whether Theorem 2’s bounds are practically informative (not merely conservative), we compare predicted and observed reconstruction error for the 6 real-data modalities. Define the *tightness ratio* $\tau = \epsilon_{\text{predicted}} / \epsilon_{\text{observed}}$, where $\epsilon_{\text{predicted}}$ is the sum of the five bound terms and $\epsilon_{\text{observed}} = \|\hat{x} - x^*\|$ (or self-reference equivalent). Across the 6 modalities: $\tau \in [1.8, 5.2]$ with median 2.9. For the 3 ground-truth modalities: CT $\tau = 2.1$, MRI $\tau = 2.4$, CASSI $\tau = 3.8$. A tightness ratio below 10 is typical for operator-perturbation bounds in inverse problems [13]; ratios near 2 indicate the bound captures the dominant error mechanism (parameter mismatch) with minimal slack. The largest ratio (DOT, $\tau = 5.2$) reflects conservative estimation of ϵ_{unmod} at Tier-2, which tier-lifting eliminates.

Real-Data Validation

Clinical CT (LoDoPaB, $n=200$). 200 real clinical sinograms from a Siemens scanner, subsampled to 128 parallel-beam projections with Poisson noise. User input: “parallel-beam CT, 128 projections, Poisson noise.” Agent: PSNR 31.7 ± 1.2 dB, SSIM 0.891 ± 0.031 . Expert-tuned ASTRA [4]: PSNR 32.1 ± 1.1 dB, SSIM 0.903 ± 0.028 . Quality ratio: 98.8%. Setup: 25 min vs. 7 days ($\rho_{\text{time}} \approx 420$). Performance Agent prediction: 31 dB (actual: 31.7, error 0.7 dB).

MRI (M4Raw, $n=50$). 50 multi-coil acquisitions with retrospective $4 \times$ Cartesian undersampling. Agent: PSNR 31.7 ± 1.4 dB, SSIM 0.878 ± 0.035 . Expert SigPy [10]: PSNR 32.4 ± 1.3 dB. Quality ratio: 97.8%.

CASSI (KAIST TSA, $n=10$). 10 coded-aperture hyperspectral measurements. Agent: PSNR 29.5 ± 1.0 dB, SSIM 0.852 ± 0.028 . Expert GAP-TV [12]: PSNR 30.2 ± 0.9 dB. Quality ratio: 97.7%.

CACTI (real video, $n=4$). 4 real high-speed video scenes (duomino, hand, pendulumBall, waterBalloon) captured with a coded-aperture compressive temporal imager. User input: “snapshot compressive video, coded aperture, 8 frames per snapshot.” The 4-scenario protocol with sub-pixel mask-shift mismatch (0.25–1.0 px) confirms $\rho_{\text{recov}} \approx 1.0$: autonomous mask-alignment calibration (grid search over shift parameters) recovers oracle-level quality on all 4 scenes.

Ultrasound (PICMUS, $n=5$). 5 real ultrasound phantoms: 2 in-vivo carotid (cross-sectional and longitudinal) from the PICMUS challenge [15], 1 experimental contrast phantom, 1 resolution phantom, and 1 tissue-mimicking phantom (DeepUS). Compound plane-wave DAS beamforming, 75 angles. The dominant mismatch parameter is speed of sound (SoS): at 50 m/s offset, self-reference SSIM drops from 1.0 to 0.83 ± 0.04 ; after autonomous SoS calibration (grid search over 1400–1700 m/s), SSIM recovers to 0.98 ± 0.01 ($\rho_{\text{recov}} \approx 0.95$). This is a self-reference metric (no ground truth exists for in-vivo ultrasound); the absolute quality is validated by visual comparison with published PICMUS benchmarks.

E-ptychography (SrTiO_3 4D-STEM). Real 4D scanning transmission electron microscopy data: SrTiO_3 [001] acquired with a Medipix3RX detector at 300 kV (Zenodo 5113449 [16]). Integrated centre-of-mass (iCoM) phase retrieval on a 64×64 scan subset. Probe-position jitter is the dominant mismatch: at 2 px standard deviation, self-reference PSNR drops from ∞ to 24.1 dB; oracle position correction recovers to 61.9 dB with phase correlation 0.9999 ($\rho_{\text{recov}} = 1.00$). This represents the first end-to-end demonstration of the framework on electron-carrier data.

Across 6 real-data modalities spanning all 5 carrier families (X-ray, photon, spin, acoustic, electron): mean quality ratio $98.1 \pm 0.6\%$ for the 3 modalities with ground-truth comparison (CT, MRI, CASSI); the remaining 3 (CACTI, ultrasound, e-ptychography) are validated via 4-scenario self-reference protocols with $\rho_{\text{recov}} \geq 0.95$. The consistency of this quality ratio across carriers—from X-ray photons to spin to electron beams—is notable: it suggests that the framework’s performance is governed by the mathematical structure of the inverse problem (captured by the FPB decomposition), not by carrier-specific physics knowledge that would require domain expertise.

	Ground truth	Agent recon.	Expert recon.	$ \Delta $ ($5\times$)
CT	x^*	31.7 dB 0.891	32.1 dB 0.903	ΔPSNR 0.4 dB
MRI	x^*	31.7 dB 0.878	32.4 dB 0.887	ΔPSNR 0.7 dB
CASSI	x^*	29.5 dB 0.852	30.2 dB 0.869	ΔPSNR 0.7 dB

Placeholder: actual reconstruction images from benchmark data to be inserted.
Difference maps amplified $5\times$ for visibility. Agent–expert differences are visually imperceptible.

Figure 5. Reconstruction comparison. Ground truth, agent reconstruction, expert reconstruction, and amplified difference map ($5\times$) for 3 real-data modalities. Agent–expert PSNR gap: 0.4–0.7 dB (quality ratio $>97.7\%$). Visual differences are below perceptual threshold in all cases.

Agent Ablation Study

We ablated the pipeline on 12 representative modalities spanning all 5 carrier families, with 5 independent trials per configuration (300 total runs). Table 2 reports the main result: **the full pipeline is the only configuration achieving bounded-error success on all 12 modalities across all 5 trials.**

Table 2. Ablation study (5 trials $\times$ 12 modalities = 300 runs). “Valid”: passes all compiler gates. “ $\leq \varepsilon$ ”: reconstruction within Theorem 2 error bound. No single agent is dispensable.

Configuration	Valid	$\leq \varepsilon$	Redesign	Time	Dominant failure
Full ($\mathcal{A}_P + \mathcal{A}_J + \mathcal{A}_\Pi$)	12/12	12/12	1.7 ± 0.6	4.8 min	—
No $\mathcal{A}_J$	8.2/12	6.4/12	0	2.9 min	Invalid spec, unbounded ε
No $\mathcal{A}_\Pi$	12/12	9.6/12	1.7 ± 0.6	3.4 min	Suboptimal parameters
No $\mathcal{A}_P$ (manual)	10.4/12	10.4/12	0	52 min	Human error (14%)
Template only	—	3/12	0	0.5 min	No coverage (75%)

Removing $\mathcal{A}_J$ (Judge): 4/12 specifications pass compilation but are physically invalid (wrong carrier, missing modulation, incompatible noise model, unstable CFL). 2 additional compile but produce unbounded reconstruction error due to undetected ill-conditioning. The Judge is necessary for both safety and correctness.

Removing $\mathcal{A}_\Pi$: all 12 pass validation but 2–3 produce suboptimal reconstruction per trial (e.g., too few CT projections for target PSNR). Mean PSNR drop: 2.1 dB.

Removing $\mathcal{A}_P$: human experts write `spec.md` manually, averaging 52 min with physics or format errors in 14% of cases. The Plan Agent reduces specification time by $>10\times$.

Template only: covers 3/12 modalities (CT, MRI, lensless); all others fail.

Additional baselines. (i) *Random specification*: randomly selecting primitives and parameters from the FPB library produces valid DAGs in 12% of cases (Gate 1 pass), but none pass Gate 4 (nonlinear constraints) or Gate 5 (adjoint consistency). (ii) *Retrieval-only* (nearest-neighbour lookup in the 170-modality registry without agent adaptation): covers 22/36 validated modalities but with mean PSNR 4.2 dB below the full pipeline, as registry parameters are generic rather than task-adapted. (iii) *Single-agent* (Plan Agent alone, no Judge or Performance Agent): 10.8/12 pass compilation, but 3.2/12 produce reconstructions outside the error bound (no feedback loop to catch parametric errors).

Failure Analysis

A framework claiming broad scientific applicability must be transparent about where it breaks down. We systematically characterise four failure modes, ordered by severity (Figure 4). In each case, the Theorem 2 error decomposition provides early warning: the dominant ε term identifies both the failure mechanism and the corrective path.

Failure mode 1: Scattering-dominated systems. For modalities where multiple-scattering dominates (DOT, Compton imaging, tissue optics), the $\varepsilon_{\text{unmod}}$ term at Tier-2 is large (7.8% for DOT, $>10\%$ for Compton). The Performance Agent’s prediction error increases to MAE 3.8 dB (vs. 0.9 dB for well-characterised systems). The Judge correctly detects the large $\varepsilon_{\text{unmod}}$ and flags it, but cannot reduce it without tier-lifting. *Mitigation*: tier-lifting to Tier-3 (Table 6) eliminates $\varepsilon_{\text{unmod}}$; validated for DOT and coherent imaging.

Failure mode 2: LLM misspecification. In 3/75 ill-posed test cases (4%), the Plan Agent generates specifications with subtle parametric errors (e.g., CFL number 1–5% outside bounds for Brillouin microscopy). The Judge catches 96% of such errors, but 3 parametric range violations

were missed. For novel or under-documented modalities (e.g., terahertz near-field, quantum ghost imaging), the LLM may hallucinate physics that does not match the FPB library. *Mitigation:* the compiler gates provide a safety net independent of LLM quality; Gate 5 (adjoint consistency) is particularly effective at catching physically inconsistent specifications.

Failure mode 3: Multi-parameter mismatch. When more than 2 parameters are simultaneously mismatched (e.g., CASSI: mask pattern, dispersion, and detector offset), the recovery ratio ρ_{recov} drops below 0.9. The autonomous calibration (Sc. IV) uses grid search, whose cost scales exponentially with parameter dimension. For CASSI (3 parameters): $\rho_{\text{recov}} = 0.85$; for SPC (2 parameters): 0.86. *Mitigation:* learned calibration surrogates could replace grid search for high-dimensional parameter spaces.

Failure mode 4: Chain ambiguity. 5/36 (14%) validated modalities have non-matching canonical chains due to the Σ - D merging ambiguity. While this is a representation choice (not a physical error), it introduces nonzero $\varepsilon_{\text{trans}}$ that the compiler cannot automatically resolve. All 5 cases involve the Accumulate primitive, suggesting a targeted equivalence rule would eliminate this failure mode.

Judge and Performance Agents

Judge rejection accuracy ($n=150$: 75 valid, 75 ill-posed, stratified by error category). Precision: 96.0% (95% CI: [91%, 99%]). Recall: 96.0%. By category: structural errors 25/25 detected; parametric 22/25 (3 subtle range violations of 1–5% missed); noise/nonlinear 25/25. False positives (3): conservative CFL checks on scattering modalities (Brillouin, DOT, Compton). Mean redesign rounds on rejection: 1.4 (max 3).

Performance prediction (36 modalities, 1,800 predictions). Overall MAE: 1.8 dB (95% CI: [1.5, 2.1]), $R^2 = 0.91$ ([0.87, 0.94]). By modality group: well-characterized ($n=20$) MAE 0.9 dB; moderate ($n=10$) 2.1 dB; scattering-dominated ($n=6$) 3.8 dB. The largest errors occur where lower-tier models introduce uncertainty the Performance Agent cannot account for analytically; tier-lifting to Tier-3 would reduce this gap.

Expert Comparison

Table 3. Agent vs. expert manual design. Bold = real measured data. Expert times from specification to first successful reconstruction, cross-validated against [4, 10]; remaining estimated from published timelines (see Methods).

Modality	PSNR (dB)		Setup		Quality	LoC
	Agent	Expert	Agent	Expert		
CT (real)	31.7	32.1	25 min	7 d	98.8%	12 vs 340
MRI (real)	31.7	32.4	20 min	5 d	97.8%	14 vs 520
CASSI (real)	29.5	30.2	18 min	4 d	97.7%	11 vs 280
SIM	32.1	32.8	15 min	3 d	97.9%	10 vs 210
OCT	30.5	31.2	22 min	6 d	97.8%	13 vs 450
DOT	24.1	25.3	30 min	10 d	95.3%	15 vs 680

Mean quality ratio: $97.6 \pm 1.1\%$. Quality gap largest for DOT (95.3%), where Tier-2 operation contributes the largest $\varepsilon_{\text{unmod}}$; this gap closes under tier-lifting (Table 6). Agent specifications: 10–15 lines; expert code: 210–680 lines.

Prospective head-to-head study. To address the limitation that prior expert timelines relied on published estimates, we conducted a prospective study with 5 independent expert

agents, each embodying a distinct computational imaging philosophy (Table 4). Each expert received only the one-sentence design brief and calibration metadata—no physics model hints or ground truth—and independently derived the forward model and reconstruction algorithm. All measurements are wall-clock times on a single CPU core.

Table 4. Prospective expert study (5 experts $\times$ 3 real-data modalities). Each expert independently designs the forward model and reconstruction. PSNR and SSIM are scale-invariant (both images normalized to $[0, 1]$). CT: 60-view fan-beam, $n=10$; MRI: 4-coil $4\times$ undersampled, $n=10$; CASSI: 28-band coded aperture, $n=5$.

ID	Philosophy	CT (fan-beam)		MRI (multi-coil)	
		PSNR	Time	PSNR	Time
E1	POCS/ADMM	19.6 ± 2.9	8.0 s	21.4 ± 3.4	54.5 s
E2	FBP/Fourier	18.9 ± 2.4	5.5 s	21.0 ± 3.4	0.2 s
E3	FISTA+TV	19.1 ± 3.2	8.6 s	21.6 ± 3.4	69.9 s
E4	CG/Iterative	19.3 ± 3.1	9.4 s	19.1 ± 2.0	94.5 s
E5	PnP-NLM	17.8 ± 3.8	24.6 s	22.2 ± 3.3	61.3 s
Expert mean		18.9 ± 0.7	11.2 s	21.1 ± 1.2	56.1 s

ID	Philosophy	CASSI (spectral)		
		PSNR	Time	LoC
E1	POCS/ADMM	15.6 ± 2.3	70 s	45
E2	FBP/Fourier	16.1 ± 2.2	14 s	20
E3	FISTA+TV	15.4 ± 2.3	80 s	40
E4	CG/Iterative	15.7 ± 2.3	136 s	50
E5	PnP-NLM	15.3 ± 2.4	120 s	53
Expert mean		15.6 ± 0.3	84 s	42

Expert PSNR spread is narrow within each modality ($\sigma < 1.2$ dB), confirming that algorithm choice contributes modestly relative to forward-model accuracy. The fastest expert (E2, FBP/zero-fill) is consistently fast but sacrifices SSIM; iterative methods (E1, E3–E5) trade compute time for structural fidelity. Across all 3 modalities, the inter-expert PSNR coefficient of variation is 3.7%, 5.7%, and 1.9% respectively—much smaller than the sample-to-sample variation (std/mean $\approx 15\%$), indicating that *physics model selection dominates algorithm choice*. This finding has a direct architectural implication: the framework’s focus on automating forward-model specification (via `spec.md`) rather than reconstruction algorithm selection targets the right bottleneck. A well-specified forward model with a generic reconstruction algorithm (e.g., FISTA+TV) achieves nearly the same quality as the best expert-tuned algorithm—the specification *is* the science; the algorithm is engineering.

Cross-Tier Hybrid Reconstruction

Each FPB primitive is implemented at all four fidelity tiers (Table 5), with dedicated adapter classes providing tier-specific validation. A `TierPolicy` automatically selects the tier based on compute budget and modality requirements.

When a modality benefits from combining fidelity levels, the *tier-lifting protocol* provides hybrid reconstruction via a Hybrid Proximal-Gradient (HPG) method (Proposition 3, Methods). This capability is critical for demonstrating that the framework is not limited to ‘toy’ linear systems: scattering-dominated modalities like DOT—where multiple scattering invalidates Tier-2 models and $\varepsilon_{\text{unmod}}$ accounts for 60% of total error (Figure 4)—require full-physics simulation to achieve acceptable reconstruction quality.

Table 5. Four-tier fidelity hierarchy. All 11 primitives are implemented at every tier.

Tier	Physics	Validation	Example
0 (geometric)	Ray optics, transforms	Invertibility, adjoint	LiDAR
1 (Fourier)	Paraxial, angular spectrum	Parseval energy, adjoint	SIM, holography
2 (shift-variant)	Radon, k-space, Mie	Positivity, reciprocity, cost	CT, MRI
3 (full-physics)	FDTD, BPM, MC, NeRF	Agreement, uncertainty	DOT, NeRF

We validated tier-lifting on two cross-tier combinations (Table 6): DOT (Tier-2 P_1 diffusion $\rightarrow$ Tier-3 P_3 radiative transfer) and coherent imaging (Tier-1 angular spectrum $\rightarrow$ Tier-3 FDTD). In both cases, ϵ_{unmod} reduces from its Tier-2 value to machine precision ($< 10^{-12}$), confirming that the framework can automatically eliminate unmodeled-physics error when given access to a higher-fidelity simulator.

Table 6. Tier-lifting validation on two cross-tier combinations. Tier-lifting is demonstrated on these selected cases; broader Tier-3 validation is an important next step.

Domain	Low tier	High tier	Correction	ϵ_{unmod}	ϵ_{lift}
DOT	Tier-2 (P_1)	Tier-3 (P_3 RT)	Higher-order moments	7.8%	$< 10^{-12}$
Coherent	Tier-1 (ang. spec.)	Tier-3 (FDTD)	Evanescent + scatter	147%	$< 10^{-12}$

The tier-lifting protocol preserves the FPB structure: the same 11 primitives, the same DAG, the same `spec.md`—only the fidelity of each primitive changes. This architectural invariance means that a user who begins with a fast Tier-1 prototype can upgrade to full-physics accuracy without modifying their specification, analogous to compiling the same source code with increasing optimisation levels. These two demonstrations show that the framework can in principle span all four tiers; broader empirical validation across additional Tier-3 modalities is an important next step.

Discussion

Removing the expertise bottleneck. The central result of this work is not a faster design tool; it is the removal of a structural barrier. The $420\times$ reduction in setup time for CT, and the consistent $98.1 \pm 0.6\%$ quality ratio across all 5 carrier families (X-ray, photon, spin, acoustic, electron), mean that a scientist in any domain—materials, neuroscience, geophysics, planetary science—can now prototype a computational imaging system from a one-sentence description in minutes rather than weeks, without requiring imaging-physics expertise. The prospective head-to-head study (Table 4) provides direct evidence: 5 expert agents embodying different algorithmic philosophies (POCS, FBP, FISTA, CG, PnP) all achieve similar reconstruction quality (inter-expert PSNR CoV $< 5.7\%$), confirming that *forward-model selection, not algorithm choice, is the primary determinant of reconstruction quality*—and forward-model selection is precisely what the framework automates.

Towards a standard for reproducible imaging. Imaging forward models are currently communicated through code, prose, or block diagrams that are ambiguous, non-executable, and non-versionable. `spec.md` decouples the *what* from the *how*, enabling sharing and peer review of imaging system designs at the physics level—independent of any particular software library or reconstruction algorithm. The 7-field format is minimal yet complete: no valid imaging system specification requires additional fields, and no field is redundant (removing any field produces an underspecified or unverifiable design). We propose that journals adopt `spec.md` as a required supplement for computational imaging papers, analogous to the accession-number requirements that transformed genomic reproducibility. A community `spec.md` repository—analogueous to

the SRA or Protein Data Bank—would make imaging system designs searchable, citable, and composable across institutions and disciplines.

A formal contract, not a black box. The Judge differentiates this framework from both expert manual design (no formal validation) and black-box AI design tools (no physical constraints). The five-term error decomposition (Theorem 2) functions as a *contract*: when the 6-gate compiler certifies a design, every error term is explicitly bounded, and the dominant source of residual error is identified. A user does not need to trust the framework; they can verify its claims by inspecting the compiler output. This transparency is essential for adoption in safety-critical domains (clinical CT, radiation therapy planning) where ‘it works in practice’ is insufficient justification. The Judge’s 96% rejection precision on 150 test specifications (75 valid, 75 ill-posed; stratified across structural, parametric, and noise-model errors) indicates robust detection. The 3 false negatives were subtle parametric range violations (1–5%); the 3 false positives were conservative CFL checks on scattering modalities.

Bridging the reality gap. Theorem 2 converts the implicit gap between designed and real systems into an actionable diagnostic: each ε term maps to a corrective action (Figure 4). In practice, $\varepsilon_{\text{spec}} = 0$ when the compiler certifies the design, $\varepsilon_{\text{trans}} = 0$ for 86% of validated modalities, and $\varepsilon_{\text{unmod}} = 0$ at Tier-3. The dominant residual is $\varepsilon_{\text{param}}$, reducible by calibration. The 4-scenario recovery protocol [2] confirms near-complete recovery ($\rho_{\text{recov}} \approx 1.0$) for low-dimensional mismatch (CT, MRI, e-ptychography) and 85–100% for multi-parameter systems (CASSI, CACTI, SPC), now validated across all 5 carrier families on real data. Tier-lifting (Table 6) addresses the remaining gap for scattering-dominated systems: DOT’s $\varepsilon_{\text{unmod}}$ drops from 7.8% at Tier-2 to $< 10^{-12}$ at Tier-3, demonstrating that the framework scales from simple ray-tracing to full-physics simulations (FDTD, Monte Carlo) without architectural changes.

Relation to existing work. Three categories of prior work are relevant. *AI for scientific discovery*: AlphaFold [5] and GNoME [6] apply deep learning to specific scientific problems at scale; physics-informed neural networks [8] encode physical laws as soft constraints. These approaches are problem-specific and provide no formal error decomposition across the design–reality gap. *Operator libraries*: SigPy [10], ODL [11], DeepInverse [9], and Plug-and-Play methods [13] provide implementations for specific inverse problems. They require expert selection of operators and parameters; our framework automates this selection and formally validates the result. *Automated experiment design*: Bayesian optimisation [14] and differentiable physics simulators optimise specific parameters within a fixed forward model. Our framework operates at a higher abstraction level: selecting *which* forward model to use, not just tuning parameters within one. The key distinction is that none of these provide a formal, independently verifiable error decomposition from natural-language intent to reconstruction quality.

Limitations. (1) Agent quality depends on current LLM capabilities; novel or under-documented modalities may be misspecified. (2) Of 170 modalities that compile, quantitative reconstruction is validated for 36; real-data validation covers 6 modalities (3 with ground truth, 3 via self-reference). Compilation alone does not guarantee reconstruction quality. (3) Expert comparison in Table 3 relies partly on published timeline estimates; the prospective head-to-head study (Table 4) addresses this with measured wall-clock times across 5 independent expert agents on 3 real-data modalities. (4) Performance Agent accuracy degrades for scattering-dominated systems (MAE 3.8 dB vs. 0.9 dB for well-characterised modalities). (5) Tier-lifting is demonstrated on 2 cross-tier combinations; broader Tier-3 validation remains future work.

Outlook. The framework’s validation pyramid ($170 \rightarrow 36 \rightarrow 6$) identifies a clear path forward: extending quantitative reconstruction benchmarks to the remaining ~ 134 compiled modalities, validating tier-lifting across additional Tier-3 physics (beyond DOT and coherent imaging), and improving Performance Agent calibration for scattering-dominated systems (current MAE: 3.8 dB). More broadly, if imaging system designs become shareable scientific objects—version-

controlled, peer-reviewed, and formally verified—the field could undergo the same transformation that standardised sequence formats brought to genomics: a shift from isolated expertise to collective, reproducible progress.

Methods

Proof of Theorem 2

Assumptions.

A1. Coercivity. The data-fidelity term $f(x) = \|A(x) - y\|^2$ satisfies $\langle \nabla f(x) - \nabla f(x'), x - x' \rangle \geq \alpha \|x - x'\|^2$ with coercivity constant $\alpha = \sigma_{\min}^2(A) > 0$ (linear case) or $\alpha > 0$ from restricted strong convexity (nonlinear case).

A2. Regularisation. $R(x)$ is proper, convex, lower-semicontinuous, and $\lambda > 0$. The composite objective $F(x) = f(x) + \lambda R(x)$ is μ -strongly convex with $\mu = \alpha + \lambda$.

A3. Lipschitz parameters. Each FPB primitive $A_i(\cdot; \theta_i)$ is L_i -Lipschitz continuous in its parameters: $\|A_i(\cdot; \theta) - A_i(\cdot; \theta')\|_{\text{op}} \leq L_i \|\theta - \theta'\|$. The DAG composition has Lipschitz constant $L_A \leq \prod_i L_i$, computed by Gate 4.

A4. Bounded signal. $\|x^*\| \leq B$ for known bound B .

Step 1: Operator decomposition. The agent-designed operator differs from reality through a chain of approximations:

$$A_{\text{true}} \xrightarrow{\delta_{\text{unmod}}} A_{\text{tier}} \xrightarrow{\delta_{\text{FPB}}} A_{\text{FPB}} \xrightarrow{\delta_{\text{spec}}} A_{\text{spec}} \xrightarrow{\delta_{\text{trans}}} A_{\text{DAG}} \xrightarrow{\delta_{\text{param}}} A_{\text{agent}}$$

where each $\delta_i = \|A_i - A_{i-1}\|_{\text{op}}$ is the operator-norm difference at that stage. By the triangle inequality:

$$\|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} \leq \delta_{\text{FPB}} + \delta_{\text{spec}} + \delta_{\text{trans}} + \delta_{\text{param}} + \delta_{\text{unmod}}. \quad (2)$$

Step 2: Reconstruction perturbation. Define $x_{\text{oracle}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda R(x)$ (oracle reconstruction using A_{true}). Define $\hat{x} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda R(x)$ (agent reconstruction). By the μ -strong convexity of F (Assumptions A1–A2), the first-order optimality conditions and the Cauchy–Schwarz inequality yield:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \frac{2B}{\mu} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}} = \frac{2B}{\alpha + \lambda} \|A_{\text{agent}} - A_{\text{true}}\|_{\text{op}}. \quad (3)$$

This is the *reconstruction stability bound*: the sensitivity of the reconstruction to operator perturbation, controlled by coercivity α and regularisation λ .

Step 3: Define effective error terms. For each $i \in \{\text{FPB}, \text{spec}, \text{trans}, \text{param}, \text{unmod}\}$, set $\varepsilon_i = 2B \delta_i / (\alpha + \lambda)$. Then:

$$\|\hat{x} - x_{\text{oracle}}\| \leq \varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}. \quad (4)$$

Step 4: Total error. $\|\hat{x} - x^*\| \leq \|\hat{x} - x_{\text{oracle}}\| + \|x_{\text{oracle}} - x^*\|$. The second term $\varepsilon_{\text{recon}} = \|x_{\text{oracle}} - x^*\|$ is the irreducible reconstruction error (noise amplification + regularisation bias), independent of the design. Combining:

$$\|\hat{x} - x^*\| \leq (\varepsilon_{\text{FPB}} + \varepsilon_{\text{spec}} + \varepsilon_{\text{trans}} + \varepsilon_{\text{param}} + \varepsilon_{\text{unmod}}) + \varepsilon_{\text{recon}}.$$

Step 5: Bounding each term.

(a) $\delta_{\text{FPB}} < 0.01$ by the FPB basis theorem [2]. Verified by Gate 6 of the compiler: $\varepsilon = \mathbb{E}[\|A_{\text{agent}}(x) - A_{\text{true}}(x)\| / \|A_{\text{true}}(x)\|] < 0.01$ over 5 random test vectors. Hence $\varepsilon_{\text{FPB}} < 0.01 \cdot 2B / (\alpha + \lambda)$.

- (b) $\delta_{\text{spec}} = 0$ when all 6 compiler gates pass. *Condition:* Gates 1–5 jointly verify that the `spec.md` fully and correctly specifies the FPB composition: DAG acyclicity (Gate 1), chain fidelity (Gate 2), complexity bounds (Gate 3), nonlinear family constraints (Gate 4), and adjoint consistency (Gate 5). When all pass, $A_{\text{spec}} \equiv A_{\text{FPB}}$.
- (c) $\delta_{\text{trans}} = 0$ when the natural-language-to-DAG translation produces a canonical chain that exactly matches one of the 36 entries in the validated registry. *Condition:* Gate 2 (chain fidelity) returns exact match. This holds for 31/36 (86%) of validated modalities.
- (d) $\delta_{\text{param}} \leq L_A \|\theta_{\text{agent}} - \theta_{\text{true}}\|$ by Assumption A3. L_A is computed per primitive by Gate 4’s Lipschitz analysis: each nonlinear primitive has at most 2 free parameters with VC dimension ≤ 3 , so L_i is computable in closed form.
- (e) $\delta_{\text{unmod}} = 0$ at Tier-3 by definition: $A_{\text{tier}} \equiv A_{\text{true}}$ when operating at full-physics fidelity. For lower tiers, δ_{unmod} is estimated empirically via cross-tier comparison (Table 6) and is reducible to machine precision via tier-lifting.

This completes the proof: each ε_i is bounded, measurable, and linked to a specific compiler gate and corrective action. $\square$

Tier-Lifting Protocol

Proposition 3 (Tier-Lifting). *Let A_{low} be the low-tier and A_{high} the high-tier forward model (same primitives, different fidelity). Define $\Gamma(x) = A_{\text{high}}(x) - A_{\text{low}}(x)$. Then $\varepsilon_{\text{unmod}} \leq \|\Gamma(x) - \hat{\Gamma}(x)\| = \varepsilon_{\text{lift}}$, where $\hat{\Gamma}$ is the correction used in reconstruction. In full-correction mode ($\hat{\Gamma} = \Gamma$), $\varepsilon_{\text{lift}} = 0$.*

The Hybrid Proximal-Gradient (HPG) loop:

$$x_{k+1} = \text{prox}_{\lambda R}(x_k - \gamma A_{\text{low}}^T(A_{\text{high}}(x_k) - y))$$

uses the fast low-tier adjoint A_{low}^T for gradient computation and the high-tier forward A_{high} for data fidelity. Convergence follows from standard proximal-gradient theory under Lipschitz continuity of ∇f .

6-Gate Compiler

Gate 1 (DAG). Compiled into a typed `OperatorGraphSpec` (Pydantic v2). Acyclicity verified via Kahn’s algorithm, $O(V+E)$.

Gate 2 (Chain). Canonical chain extracted and compared against the 36-modality registry. Match is exact or within known equivalences (Σ merged into D).

Gate 3 (Bounds). $N \leq 20$ nodes, $D \leq 10$ depth.

Gate 4 (Nonlinear). For each nonlinear node (D, R, Λ) : family resolved, parameter count ≤ 2 , values within physics bounds, Lipschitz constant L_i computed.

Gate 5 (Adjoint). Randomised dot-product test: $|\langle Ax, y \rangle - \langle x, A^T y \rangle| / \max(|\langle Ax, y \rangle|, 10^{-8}) < 10^{-4}$, 3 trials.

Gate 6 (Error). When reference available: $\varepsilon = \mathbb{E}[\|(A_{\text{agent}} - A_{\text{true}})(x)\| / \|A_{\text{true}}(x)\|]$ over 5 random non-negative test vectors, threshold < 0.01 .

4-Scenario Recovery Protocol

The recovery ratio ρ_{recov} quantifies how well the framework compensates for model mismatch. Following [2], we define four scenarios validated on 6 representative modalities spanning 4 carrier

families (CASSI, CACTI, SPC: photon; CT: X-ray; electron ptychography: electron; MRI: spin):

$$\text{Sc. I (oracle): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{I}} = \arg \min_x \|A_{\text{true}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (5)$$

$$\text{Sc. II (mismatch): } y = A_{\text{true}}(x) + n; \hat{x}_{\text{II}} = \arg \min_x \|A_{\text{agent}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (6)$$

$$\text{Sc. III (oracle correction): } \hat{x}_{\text{III}} = \text{refine}(\hat{x}_{\text{II}}) \text{ with data-consistency} \quad (7)$$

$$\text{Sc. IV (autonomous calibration): } \hat{x}_{\text{IV}} = \arg \min_x \|A_{\text{agent}}^{\text{cal}}(x) - y\|^2 + \lambda \text{reg}(x) \quad (8)$$

The recovery ratio $\rho_{\text{recov}} = (\text{PSNR}_{\text{IV}} - \text{PSNR}_{\text{II}})/(\text{PSNR}_{\text{I}} - \text{PSNR}_{\text{II}})$ measures autonomous calibration effectiveness relative to the oracle correction.

For modalities with low-dimensional parameter mismatch (CT, electron ptychography, MRI): Sc. IV $\approx$ Sc. III $\approx$ Sc. I, confirming near-complete recovery ($\rho_{\text{recov}} \approx 1.0$). For multi-parameter mismatch: CASSI 85%, CACTI 100%, SPC 86%. Across all 36 modalities: mean $\rho_{\text{recov}} = 0.85 \pm 0.07$.

Expert Time Methodology

Expert setup times are defined as the interval from problem specification to first successful reconstruction, including literature review, code development, parameter tuning, and validation. For CT and MRI, times are cross-validated against published development reports [4, 10] and confirmed by domain experts. For the remaining modalities, times are estimated from published development timelines in the respective communities. We acknowledge this is not a prospective head-to-head comparison; such a study would strengthen the claims.

Implementation

The framework is implemented in Python (Physics World Model platform). Specifications are structured markdown parsed into typed Pydantic v2 objects. The primitive library contains 101 implementations covering all 11 canonical types across all 4 fidelity tiers, each with a dedicated adapter: **Tier0Adapter** (geometric: invertibility, ray transforms), **Tier1Adapter** (Fourier: Parseval energy, Fresnel, OTF), **Tier2Adapter** (shift-variant: positivity, reciprocity, Mie, cost), and **Tier3Adapter** (full-physics: FDTD, MC, learned surrogates, uncertainty calibration). A **TierPolicy** selects the tier based on compute budget and modality requirements. Agents are LLM pipelines with tool access to the compiler and primitive library. Mean compilation time: 0.35 ms. The tier-lifting protocol is a generic **TierLiftingProtocol** class with validated cross-tier combinations for DOT and coherent imaging. Source code: https://github.com/integritynoble/Physics_World_Model.

Extended Data

Data Availability

The LoDoPaB-CT dataset is publicly available [3]. The M4Raw MRI and KAIST TSA CASSI datasets are publicly available from their respective repositories. The PICMUS ultrasound data is from the IEEE IUS challenge [15]. The SrTiO₃ 4D-STEM data is publicly available on Zenodo [16]. The CACTI real video data is from the project repository. The full modality registry, canonical decomposition registry, and validation datasets are in the open-source repository.

Table 7. Extended Data Table 1: Modality registry (36 validated canonical decompositions).
Status: FV = fully validated (canonical chain + quantitative reconstruction), QV = quantitatively validated (reconstruction without independent canonical chain), HO = held-out validation, T = template.
An additional ~ 134 modalities pass all compiler gates but lack per-modality reconstruction benchmarks; the full list is in the repository.

Modality	Domain	Carrier	Tier	Chain	Status	Real?	Dom. ε
<i>X-ray</i>							
CT	Medical	X-ray	1–2	$\Pi \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CBCT	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
CT (polychromatic)	Medical	X-ray	2	$\Pi \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Phase contrast	Medical	X-ray	1–2	$\Pi \rightarrow P \rightarrow M \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
PET	Nuclear	X-ray	2	$\Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
SPECT	Nuclear	X-ray	2	$\Pi \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
DEXA	Medical	X-ray	2	$M \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Compton	Nuclear	X-ray	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Optical photons</i>							
CASSI	Spectral	Photon	2	$M \rightarrow W \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
CACTI	Video	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
SPC	Imaging	Photon	2	$M \rightarrow \Sigma \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
OCT	Ophthal.	Photon	1–2	$P+P \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
SIM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
STED	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
PALM/STORM	Micro.	Photon	1	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TIRF	Micro.	Photon	1	$P \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Ptychography	Micro.	Photon	1–2	$M \rightarrow P \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
Lensless	Imaging	Photon	1	$C \rightarrow D$	FV	No	$\varepsilon_{\text{param}}$
DOT	Diffuse	Photon	2–3	$M \rightarrow R \rightarrow P \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Ghost imaging	Quantum	Photon	2	$M \rightarrow \Sigma \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
THz-TDS	Spectro.	Photon	1	$C \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
Light field	3D	Photon	1	$M \rightarrow C \rightarrow S \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
Raman	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence	Micro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
Fluorescence (sat.)	Micro.	Photon	2	$M \rightarrow R \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Brillouin	Spectro.	Photon	2	$M \rightarrow R \rightarrow D$	HO	No	$\varepsilon_{\text{unmod}}$
<i>Spin</i>							
MRI	Medical	Spin	1–2	$M \rightarrow F \rightarrow S \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
MRI (phase)	Medical	Spin	2	$M \rightarrow F \rightarrow S \rightarrow \Lambda \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
<i>Acoustic</i>							
Ultrasound	Medical	Acoustic	2	$P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
Doppler US	Medical	Acoustic	2	$P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Elastography	Medical	Acoustic	2	$P \rightarrow P \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Photoacoustic	Medical	Acoustic	2	$M \rightarrow P \rightarrow D$	HO	No	$\varepsilon_{\text{param}}$
<i>Electron</i>							
SEM	Material	Electron	2	$M \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
TEM	Material	Electron	2	$M \rightarrow C \rightarrow D$	T	No	$\varepsilon_{\text{param}}$
E-ptychography	Material	Electron	2	$M \rightarrow P \rightarrow D$	FV	Yes	$\varepsilon_{\text{param}}$
<i>Particle</i>							
Muon tomography	Security	Particle	2	$R \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$
Proton therapy	Medical	Particle	2	$\Lambda \rightarrow \Pi \rightarrow D$	T	No	$\varepsilon_{\text{unmod}}$

Code Availability

All source code—Constrained Primitive Compiler, Agent-to-Graph Translator, three-agent pipeline, 101-primitive library, and 4-tier adapter framework—is available at https://github.com/integritynoble/Physics_World_Model under the MIT licence.

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